

DATA ANALYTICS PROJECT



24.91M

Total Revenue

10.46M

Total Profit

1.66

YoY Revenue Growth %

0.42

Profit Margin %

17K

Total Customers

Adventure Works Business Analysis

T-Profit by SubcategoryName



Continent	Total Revenue	YoY Revenue Growth %	Notes
Pacific	7,416,455.01	1.25	
Europe	7,789,883.95	1.98	
North America	9,708,244.70	1.82	

End-to-End Business Analytics Project
Turning Data into **Strategic Insights**



SQL



Python



Power BI

Total Revenue and Total Profit by Region

● Total Revenue ● Total Profit



By **Gajendra Singh**
Data Analyst



Business Problem
Solving



Data-Driven
Insights



Interactive
Dashboards



Strategic
Decisions



Executive Context

- Global operations across products, customers, and regions
- Increasing business scale adds complexity to performance visibility
- Leadership requires a unified, data-driven view for strategic decisions



Problem Statement

- Revenue and profit are unevenly distributed across products and customers
- Growth varies significantly across regions and time periods
- High dependency on a limited set of products and customer segments
- Returns reduce profitability and introduce operational risk

Project Overview



Business Performance

Revenue, profit, growth trends and operational scale analysis.



Product Strategy

Category contribution, top products and portfolio concentration.



Customer & Market

Customer segmentation and regional performance dynamics.



Risk & Forecasting

Return impact analysis and predictive revenue modeling.



Data Foundation

Enterprise Data Scope

- Multi-year transactional sales records
- Product categories and subcategories
- Customer profiles and order behavior
- Regional and country-level dimensions
- Returns impacting revenue and profit

Data Scale

- 24.9M+ Total Revenue
- 25K+ Orders
- 17K Customers

Data Integration

- Sales linked to Products
- Sales linked to Customers
- Sales linked to Returns
- Time-based performance tracking

Business Performance Insights

Key Observations

- The business operates at significant scale with strong overall revenue and profit
- Growth exists but remains inconsistent across time periods
- Performance varies across regions and product segments
- Profitability levels indicate operational and cost variability

Business Meaning

- Sustained growth depends on managing variability across markets
- Performance stability requires better alignment of demand, pricing, and costs





Dashboard Analysis & Insights

Executive Overview



24.91M

Total Revenue

10.46M

Total Profit⁺

1.66

YoY Revenue Growth %

0.42

Profit Margin %

17K

Total Customers

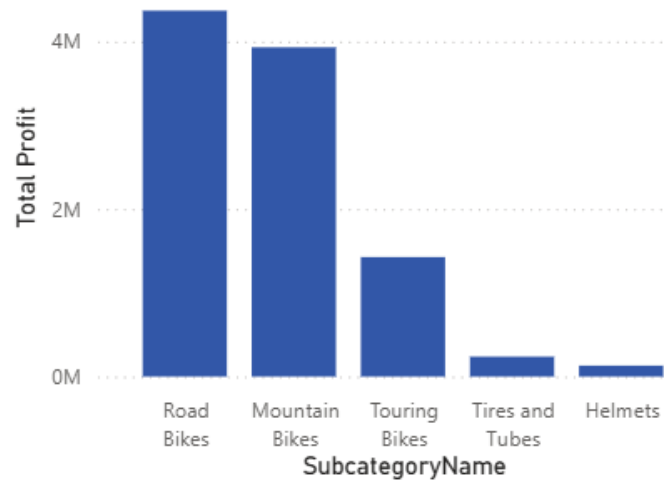
Year

All

Region

All

Total Profit by SubcategoryName

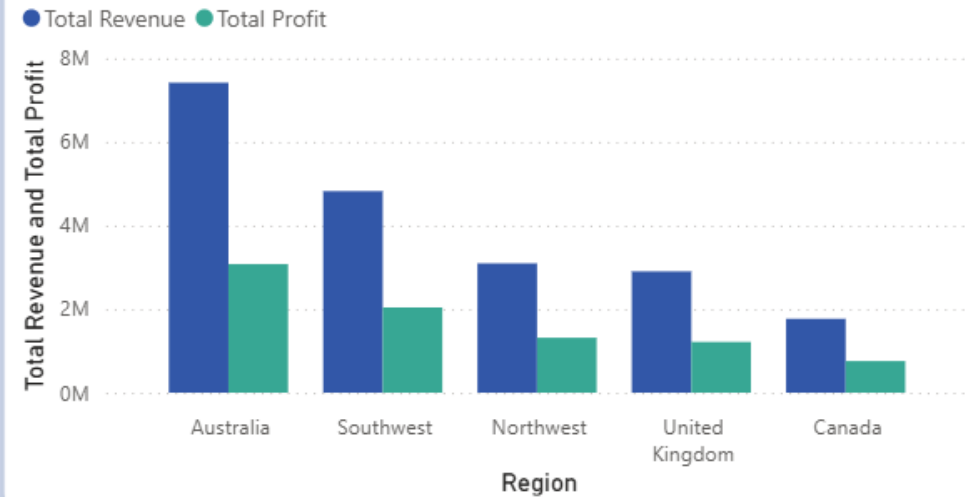


Continent	Total Revenue	YoY Revenue Growth %	Total Revenue by Month Name
Pacific	7,416,455.01	1.25	
Europe	7,789,883.95	1.98	
North America	9,708,244.70	1.82	

Total Revenue by Month Name



Total Revenue and Total Profit by Region



Revenue & Growth



1.8M

Revenue MTD

9.2M

Revenue YTD

9.36M

Revenue LY

5.1M

Revenue QTD

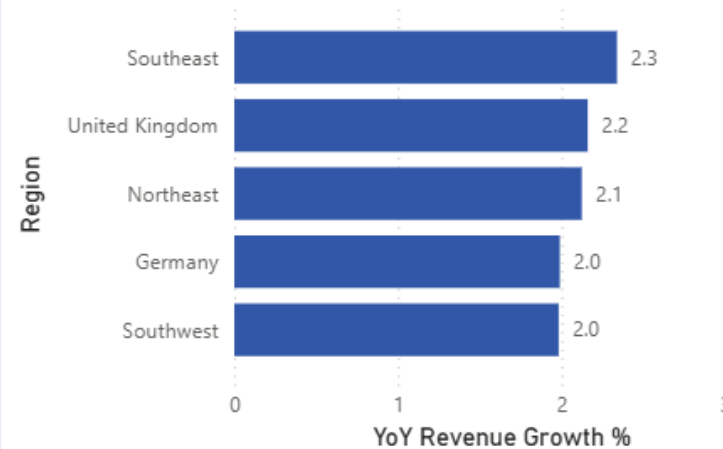
Region

All

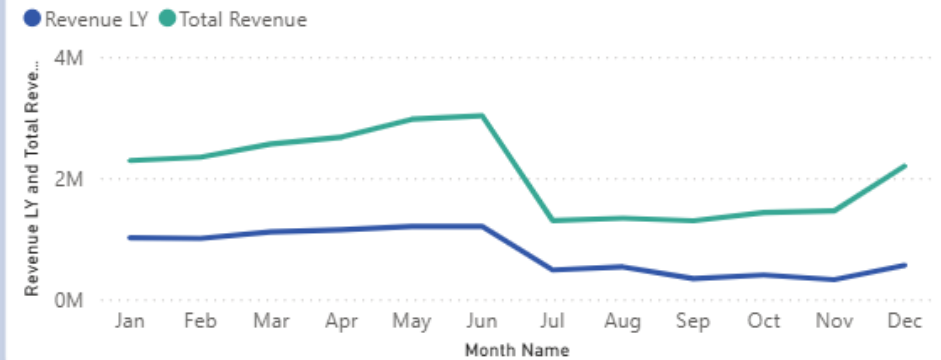
Year

All

Top 5 Regions by Revenue Growth %



Revenue LY and Total Revenue by Month Name

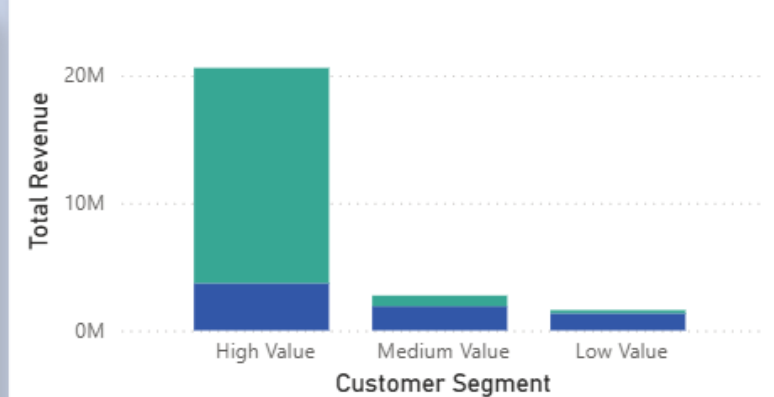


MoM Growth % by Month Year



Total Revenue by Customer Segment and Customer Type

Customer Type: New Customer (Blue), Repeat Customer (Green)



Product & Category Insights



990.1
AOV

0.56
Profit Margin %

3.89M
Profit YTD

25K
Total Orders



CategoryName

All

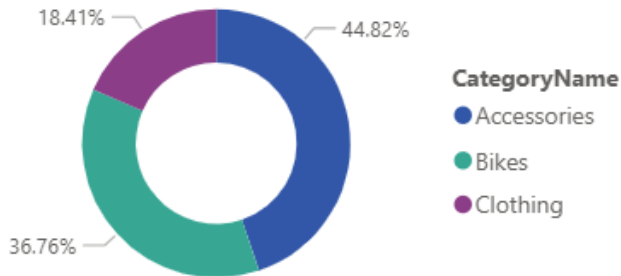
Region

All

Year

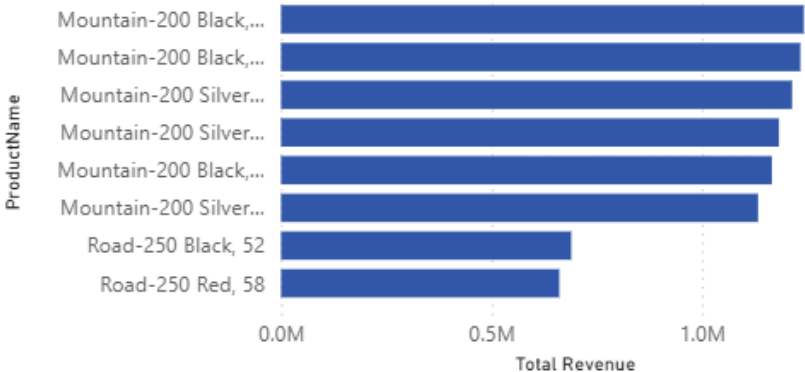
All

Total Orders by CategoryName

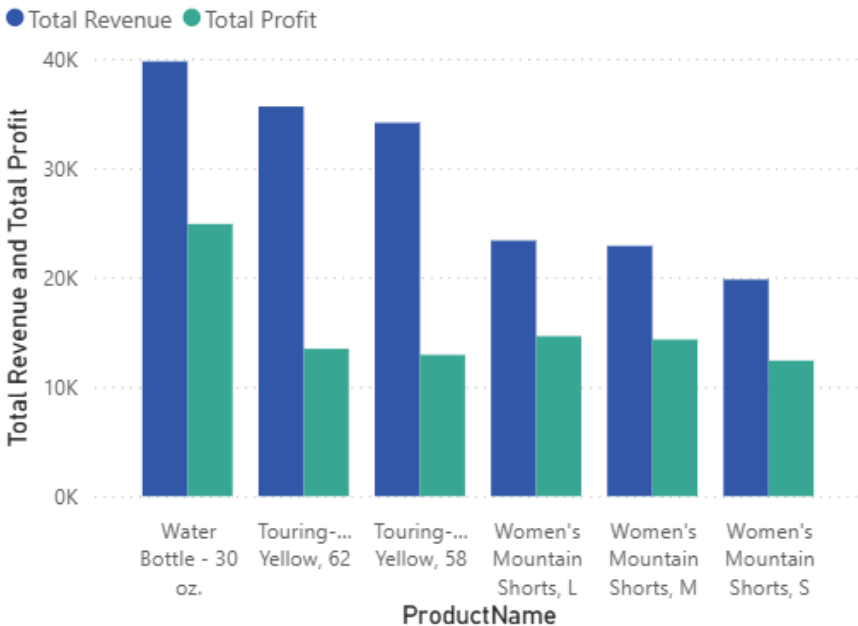


CategoryName	Total Revenue	Return Rate % 2	Total Revenue by Month Name
Bikes	23,642,500.06	0.13	
Accessories	906,656.58	0.11	
Clothing	365,427.02	0.26	

Total Revenue by ProductName



Total Revenue and Total Profit by ProductName



Customer Analytics



17K

Total Customers

1.43K

Revenue per Customer

0.87

Repeat Purchase Rate %

Customer Type

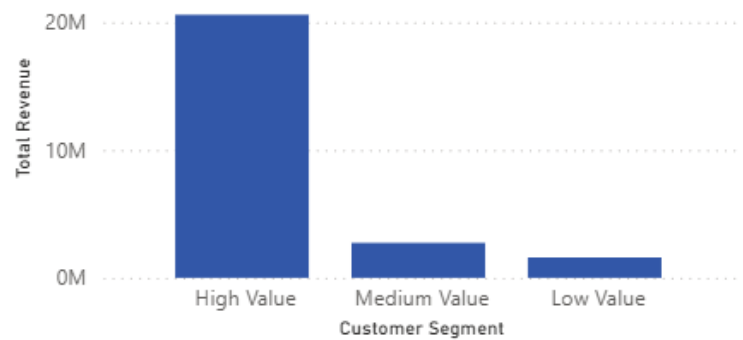
New Customer

Repeat Customer

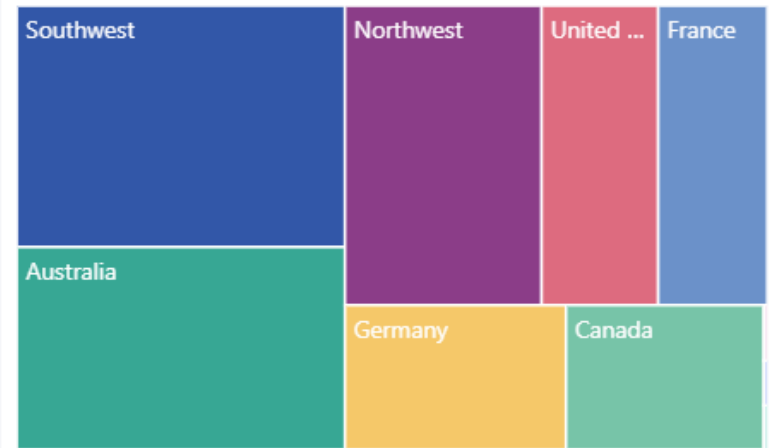
Year

All

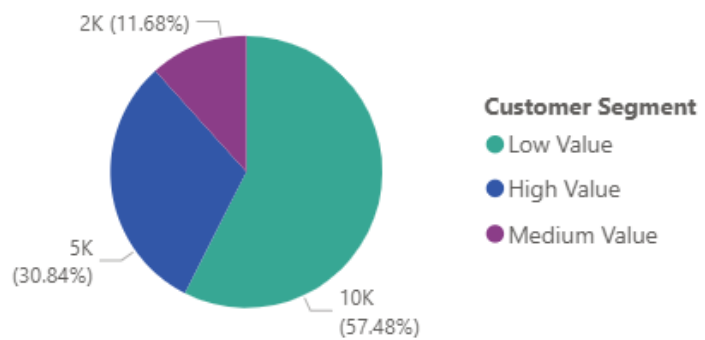
Total Revenue by Customer Segment



Customer Count by Region

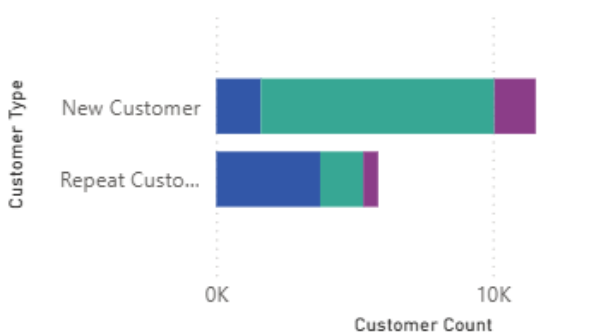


Total Customers by Customer Segment

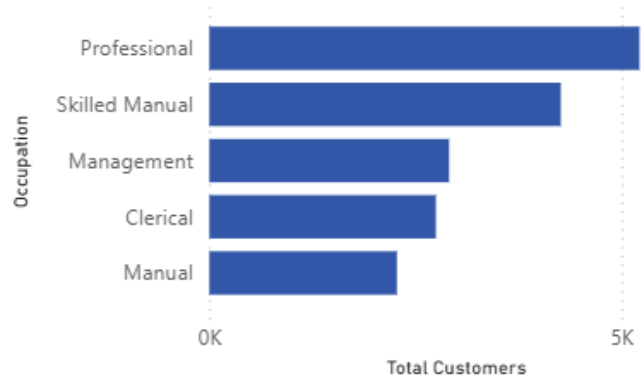


Customer Count by Customer Type and Customer Segment

Customer Segment: High Value (Blue), Low Value (Green), Medium Value (Purple)



Total Customers by Occupation



Returns & Risk Analysis



124

High Risk Products

0.07

Return Rate % 2

1828

Total Returns

Year

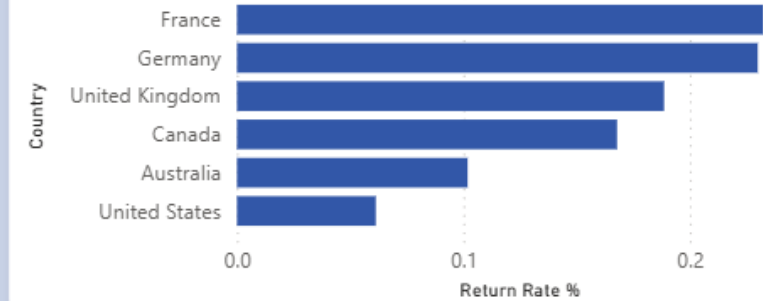
All

Region

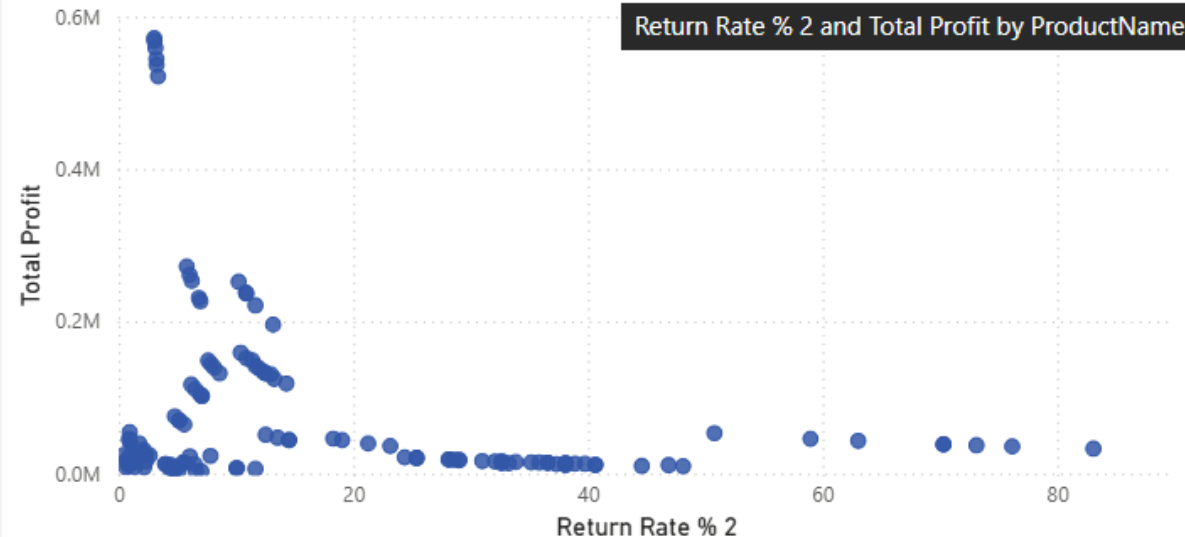
All

Country	Accessories	Bikes	Clothing
Australia	▲ 0.16	▲ 0.45	▲ 0.71
Canada	▲ 0.22	■ 2.13	■ 1.05
France	▲ 0.34	■ 1.34	■ 1.70
Germany	▲ 0.34	■ 1.23	■ 1.66
United Kingdom	▲ 0.28	■ 1.07	■ 1.28
United States	▲ 0.09	▲ 0.41	▲ 0.40

Return Rate % by Country



Return Rate % 2 and Total Profit by ProductName



Return Rate % 2 by CategoryName



Product & Portfolio Insights

Revenue Concentration Pattern

- A small group of product categories drives the majority of total revenue
- Revenue contribution drops sharply beyond top products

Profitability Imbalance

- High sales volume does not always translate into high profit
- Some products generate volume but operate on lower margins

Portfolio Structure

- Performance is highly concentrated across limited products
- Heavy dependence increases strategic vulnerability

Strategic Implication

- Portfolio optimization should balance **revenue leadership and margin strength**
- Growth opportunities exist in margin improvement and category diversification

Customer Value Insights



Revenue Concentration

- A small customer segment drives a large share of revenue
- Revenue contribution is highly uneven



Loyalty Impact

- Repeat customers deliver higher lifetime value
- One-time buyers add volume but limited long-term value



Customer Value Variance

- Spending patterns vary significantly across segments
- High-value customers show stronger purchase frequency



Strategic Direction

- Retentive growth is more valuable than acquisition alone
- Loyalty initiatives can significantly increase customer lifetime value

Regional & Market Insights



Revenue Distribution

- "North America leads with \$9.7M revenue"
- A few markets drive the majority of total sales



Market Performance Variation

- Country-level performance differs within the same region
- Strong regions may still contain underperforming markets



Growth Pattern Differences

- Revenue growth trends are not uniform across regions
- Mature markets show stability while emerging markets show growth potential



Strategic Direction

- Regional strategies must reflect market maturity and demand patterns
- Expanding in underpenetrated markets presents growth opportunities

Risk & Profitability Insights



Return Impact Risk

- Product returns create direct revenue leakage
- Returns are concentrated within a limited set of products



Margin Pressure Areas

- High sales volume does not always translate into strong profit
- Certain categories operate under lower margin structures



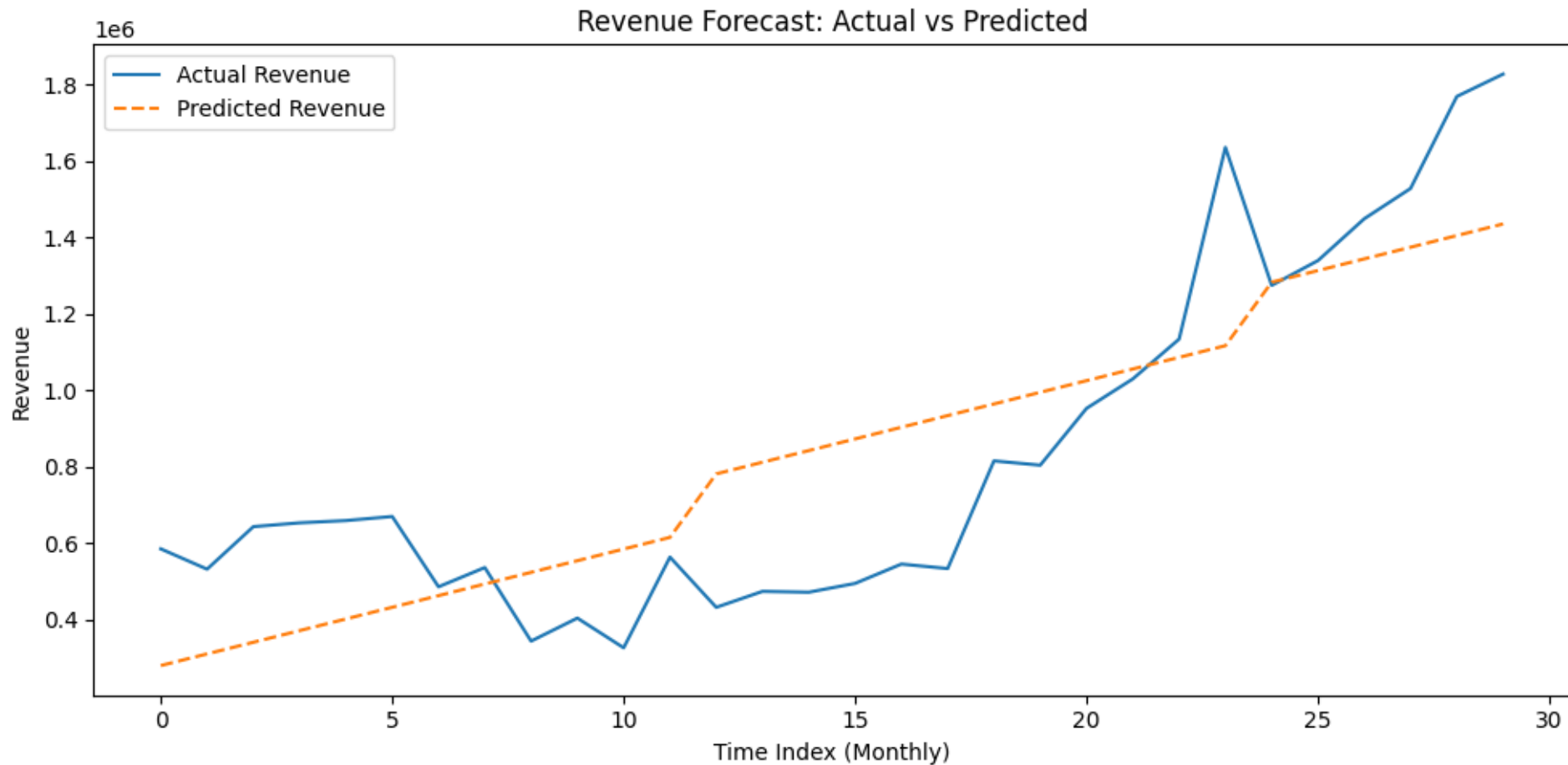
Profit Variability Drivers

- Profitability varies across products, regions, and customer segments
- Operational costs and return rates influence margin stability



Strategic Direction

- Reducing returns can significantly improve profitability
- Margin-focused product and pricing strategies can strengthen financial performance



Growth Outlook

Revenue is expected to continue rising over time.



Forecast Reliability

Predicted trend aligns with historical growth pattern.

⚠️ Variability Note

Short-term fluctuations may occur, but overall direction remains positive.

Regional Comparison

	Region	Revenue	Profitability	Growth
1	North America	\$9.7M	Strong	Stable
2	Europe	\$7.8M	Medium	Growing
3	Pacific	\$7.4M	Medium	Fast
4	LATAM	Low	Low	Emerging



Decision : Before vs After Analytics

BEFORE ANALYSIS

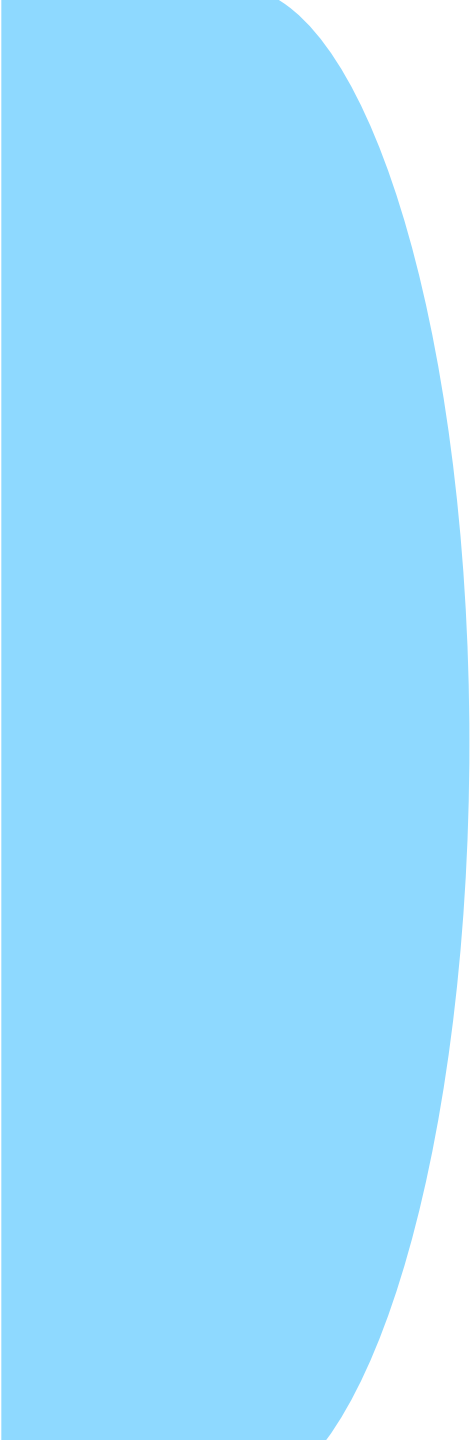
- Fragmented reporting across departments
- Limited visibility into emerging risks
- Decisions based on historical reactions
- Delayed response to performance issues

AFTER ANALYSIS

- Unified, real-time performance visibility
- Early identification of risks & trends
- Data-driven strategic decision-making
- Faster and proactive business response

Key Findings Summary

- Business scale is strong but uneven across segments
- Revenue depends heavily on limited products and customers
- Returns significantly reduce profitability
- Growth varies by season and region
- Customer retention delivers higher long-term value



Business Applications

- Monitor performance through executive dashboards
- Optimize product portfolio & pricing strategies
- Strengthen customer retention initiatives
- Improve regional marketing & promotion planning
- Reduce returns and operational cost leakage

Limitations & Considerations

- Analysis is based on historical data patterns
- External market factors were not included
- Performance trends may evolve over time
- Business impact depends on execution quality

Future Opportunities

- Advanced demand forecasting & scenario planning
- Customer churn prediction models
- Real-time performance monitoring dashboards
- Automated alerts for risk & performance changes





Conclusion

Data analytics enables Adventure Works to move from reactive reporting to proactive decision-making.

By combining performance insights with predictive analysis, the business can improve profitability, reduce risk, and support sustainable growth.

Thank you

Gajendra Singh

Let's Connect

Thank you for your time.

I'd be happy to answer any questions.

